Hackathon Guide

How to get started

1. **Set up your team repository.** Open the hackathon template, select Use this template → Create a new repository, and clone your new repository to your computer.
2. **Register your team.** Join the hackathon Slack and send one message per team to the HABS Bot with your repository link or *your-name/your-repo*. The bot will send you a team key and a results link. Register even if you do not plan to use a GPU. For GPU runs, add the team key to *.github/workflows/submit.yml* as described in the repository README. Avoid renaming or moving your repository: the key is linked to its address.
3. **Explore the data.** Read DATASET.md and try the sample recording and starting notebooks. To download data to your computer, request access to the full dataset on Hugging Face. The full dataset is already available to GPU jobs.
4. **Write and test your code.** Implement your approach in *process.py*, add dependencies to *requirements.txt*, and test locally using the commands in the repository README. You can complete your project locally if it does not require a GPU.
5. **Run on a GPU, if needed.** Commit and push your changes, then select Actions → HABS → Run workflow in your GitHub repository. Pushing alone does not start a GPU run. Jobs can run for up to one hour. Use your results link with Azure Storage Explorer to access the output and logs.

What to research

We provide a very large and rich multimodal physiological dataset of people undergoing various emotional states. We describe it in detail in the document *dataset_paper.pdf*.

What can you do with such a dataset? In general a lot. For example:

- Understand the biological foundation of *fear* to enable treatment for people with anxiety disorders
- Decode mental states from brain activity to improve AI-assisted psychotherapy
- Measure joy in real-time to adapt viewed content to the person's preference
- ...

Given how large and rich the dataset is, we do not provide concrete tasks for you. Finding a meaningful research question is part of the journey!

However, you should be aware that this dataset is complex. Even for an experienced post-doctoral researcher it would be ambitious to make good use of the full dataset in just four weeks. Therefore, less is often more. Focusing on a narrow research question in favor of going deep may be a more promising approach than aiming too high.

Exemplary research questions

To not stay too vague, we provide some exemplary research questions here:

Data Processing

- Develop an objective quantification of EEG signal quality
- Find the best preprocessing pipeline of pupillometry data
- Develop an ECG cleaning algorithm
- Engineer galvanic skin response (GSR) features strongest related to fear
- Reliably calculate heart rate variability (HRV) from ECG on the highest time resolution (shortest windows) possible
- Develop an algorithm for detecting bad EEG segments or channels
- Multimodal quality control: Build a dashboard that flags missing or unreliable periods across EEG, ECG, skin conductance, respiration and eye-tracking data, and shows which intervals are usable
- Put every sensor on the same clock: Develop a method for aligning streams with different sampling rates, missing segments and clock drift. Test its robustness by introducing known timing errors and measuring how accurately they are recovered. The expected deliverable is an alignment pipeline and a report of its remaining timing error and effects on downstream analyses. Recording offsets should be distinguished from physiological response delays.
- ....

Multimodal AI

- Predict one modality from another (e.g. predict the heart rate from the breath rate or the pupil diameter from the EEG)
- Predict self-reported arousal better from multiple signals compared to isolated signals
- Develop a representation learning method using at least two signal modalities. Optimize and evaluate the obtained representations (including downstream tasks linear probing) and compare them with prediction baselines.

- Are all signals and electrodes from each modality truly useful for predictions? Propose and develop a method capable of identifying the most and least useful signal sources and electrodes for predicting self-reported targets.
- Align information across modalities: The recordings share a clock, but the same event may appear at different times in different signals—for example, in EEG at time t and in GSR at $t + k$, where k may vary. Develop a method to learn these cross-modal correspondences and test whether they improve emotion prediction over combining signals at matching timestamps.
- …

Real-Time Deployment

- Predict the pupil diameter from the EEG in real-time
- Detect eye blinks from the EEG in real-time
- Predict video-induced emotions in real-time
- Detect and remove EEG artifacts in real-time
- A live multimodal observatory: Build a pipeline that processes incoming signals and displays synchronized features, signal quality and processing delay. Recorded data can be replayed as streams to provide a controlled development environment. The expected deliverable is a working dashboard with measurements of processing latency and computational requirements.
- Detect a change as it happens: Develop an online method for detecting changes in physiological dynamics. Evaluate how quickly it detects relevant changes and how often it produces false alarms. Protocol markers and reviewed events can provide evaluation references, while recognizing that a protocol event does not necessarily produce an immediate physiological change. The expected deliverable is an online detector with a quantified trade-off between detection delay and false alarms.
- Keep working when a sensor fails: Develop a system that remains useful when a sensor becomes noisy, delayed or unavailable. Possible approaches include weighting modalities according to their quality or switching to models that require fewer sensors. The expected deliverable is a streaming prototype tested under controlled sensor failures, showing how prediction performance and confidence change as data quality deteriorates.
- ….

You will be doing cutting edge-research! That means that these problems are very hard and we cannot guarantee that working solutions exist. If you “fail” (e.g. random decoding baseline) it may be because the problem cannot be solved. If you provide well-researched

and compelling reasons of why a better solution is not possible, you may still win the hackathon!

How we evaluate

You will be evaluated on your 1) GitHub repo, 2) recorded powerpoint presentation, and 3) written summary.

GitHub repository

The code base should be well documented, readable, and reproducible. Provide some example scripts how to replicate your most important findings. Important: The less code the better! Keep files and lines of code to a bare minimum. While you will write a lot of code during research (especially with the help of AI) your most important results can likely be expressed in just a handful of well documented python files. Document your environment well and provide 1-2 impactful README files or jupyter notebooks to highlight your most important findings.

Powerpoint presentation

While we want to learn about your results, we are also very interested in how you got there. What confused you? What took longer than expected? In which cases did AI tools help you and in which cases could you not rely on it? What was your biggest learning? What would you do if you had additional 4 weeks to work on your project?

Summary

Provide a pdf in form of a short research paper. **Abstract:** Summarize your approach, methodology, results, and interpretation in a dense abstract. **Introduction:** Briefly explain your research question. You do not need to review and re-cite the existing literature. **Methods:** Be exact in the methodology of your final submission. **Results:** Explain what you found. **Discussion:** What do you think it means? Keep it rather short to a minimum of 2 pages and a maximum of 5 pages. In addition to this page count, provide at least two figures (Hint: The figures are usually the most important part of a paper!).

Selecting the winner

The jury evaluates the quality of the research, not just whether a model achieves a high score. We look for a focused question, sound methods, evidence that supports the conclusions, useful learning, reproducible work, and clear communication. A well-supported negative result or a careful account of a limitation can be as strong as a

successful model. Teams are assessed in light of their question and the evidence available to them; model scores are compared only when the tasks and baselines are comparable.

Criterion	Weight	What judges look for
Research question and scope	15%	A clear, focused question that fits the available data and time
Method and evidence	35%	Appropriate comparisons and baselines; conclusions supported by the results
Insight and learning	25%	A useful finding, pipeline, or limitation that others can learn from
Reproducibility & Communication	25%	Enough documentation and code to follow or reproduce the main result. A clear explanation of the question, approach, findings, and limitations